PORTFOLIO PROJECT | PROJECT CHARTER + IMPLEMENTATION REPORT

Why Are Customers Receiving Orders Late?

A Data-Driven Investigation into E-Commerce Delivery Performance and Customer Satisfaction

Project roadmap • Technical design • Analytical methodology • Learning outcomes • Validation framework

DOCUMENT CONTROL	DETAIL
Version / date	v1.1 — 30 October 2026
Purpose	Portfolio-ready end-to-end data analytics case study
Source data	Olist Brazilian E-Commerce Public Dataset (2016–2018)
Core tools	Python (pandas), SQL / SQLite, Power BI, GitHub
Status	Starter pipeline implemented and tested on synthetic fixture; empirical Olist analysis pending
Estimated schedule	14 days at 3–5 hours per day (intensive sprint)

EVIDENCE INTEGRITY This is a full project design and implementation report, not a final findings report. No empirical late-delivery rates or causes are claimed before running the official Olist data.

Prepared as an independent professional development and recruitment portfolio project.

Executive summary

This project examines whether e-commerce orders reach customers by their promised delivery date, which observable fulfilment stages and segments are associated with lateness, and how delivery performance relates to customer review scores. It uses real historical order data released by Olist, plus transparent Python/SQL transformations and a planned Power BI decision dashboard.

The core analytical unit is one unique order. The starter repository includes CSV validation, aggregation of multi-item and multi-review records, a consistent late-delivery indicator, exclusion logs, descriptive investigations and executable tests. The same logic can be verified through SQL and then communicated in Power BI.

The deliverable at completion will be an evidence-led investigation: quantified baseline, segment and stage comparisons, review-score analysis, robustness checks, dashboard, recommendations, limitations and a portfolio narrative. The project does not claim to identify individual courier failures or causal mechanisms not recorded in the source data.

FIRST SUCCESS MILESTONE Run the official Olist data through the starter pipeline, audit the denominator, and reproduce the baseline late-delivery rate using independent SQL. This must happen before interpreting patterns.

1. Business case and project charter

1.1 Business context

Marketplaces depend on accurate delivery promises. Late deliveries can affect customer satisfaction, refund requests, support workload and repeat purchases. Even if overall performance is acceptable, pockets of poor performance may appear among geographical regions, high-volume sellers or fulfilment stages. Management needs a reliable measurement system before taking action.

1.2 Problem statement

Management does not yet have a validated, order-level explanation of late deliveries. Unreconciled raw joins can overcount orders, inconsistent “on-time” definitions can distort metrics, and reviewing only problematic orders can create bias. The analyst must first construct trustworthy measures and then investigate associations without making unsupported causal claims.

1.3 Objective and decision

Primary objective: measure lateness relative to the customer-facing promise and identify high-priority areas where operational investigation or controlled process improvements may be justified. Decision supported: where should operations management investigate first, what evidence supports that priority, and how should results be monitored?

1.4 Scope and exclusions

IN SCOPE	OUT OF SCOPE / REQUIRES ADDITIONAL DATA
Delivered orders and promised-vs-actual calendar date	Proving which carrier or employee caused a delay
Observed purchase/approval/handoff/delivery durations	Real-time tracking and currently open orders
Customer-state and single-seller comparisons	Reliable per-item attribution for multi-seller orders
Review-score relationships and subgroup sensitivity	Proving delays cause low reviews or lost revenue
Data-quality audits and Power BI dashboards	Verified profit impact, refund cost, or carrier SLAs

2. Research questions and hypotheses

QUESTION	MEASURE / INVESTIGATION
RQ1. How common are late deliveries?	Late / eligible delivered orders; monthly baseline
RQ2. When is a delivery most likely to become late?	Compare valid approval, pre-handoff and in-transit durations
RQ3. Which regions show higher late rates?	State-level volume and late rate with minimum sample size
RQ4. Are some sellers associated with more late orders?	Single-seller orders only; min-volume filter
RQ5. Are late orders associated with worse reviews?	Review coverage, mean score and low-score share
RQ6. How robust are the patterns?	Alternative group filters, month and geography stratification

Initial hypotheses, to be tested rather than assumed: H1, delays cluster more in some customer states than others; H2, longer observed pre-handoff or transit intervals are associated with missed promises; H3, orders arriving after the promised date receive lower review scores on average. All hypotheses may be rejected or remain inconclusive.

3. Source data and analytical limitations

Source: Olist Brazilian E-Commerce Public Dataset, published on Kaggle by Olist and collaborators (approximately 100,000 anonymised Brazilian orders from 2016 to 2018). It includes order status and lifecycle dates, items, sellers, customer geography, and reviews. It is historical data: patterns are not evidence of the current marketplace situation.

SOURCE CSV	KEY FIELDS	ROLE
olist_orders_dataset.csv	order_id, timestamps, order_status	Order lifecycle, promised vs actual dates
olist_customers_dataset.csv	customer_id, city, state	Customer-region segmentation
olist_order_items_dataset.csv	order_id, item_id, seller_id, prices	Order-level volume and distinct sellers
olist_sellers_dataset.csv	seller_id, state, city	Single-seller location comparisons
olist_order_reviews_dataset.csv	order_id, review_score	Reviewed-order satisfaction analysis

Other available files, including geolocation, products, payment methods and category translations, can support later extensions. Data provenance, download link and terms should be retained in the final repository; raw data should not be republished without reviewing its licence.

DATASET CAUTION The dataset is anonymised and contains no trustworthy carrier identity, recorded reason for delay or randomized intervention. Time between handoff and delivery is observable; the causal reason for that duration is not.

4. Data architecture and grain

Ingestion: official CSVs → Python input/schema checks → source-specific aggregation → single fact_orders table → SQLite and CSV output → SQL investigations → Power BI visuals → documented recommendations.

4.1 Prevent double counting

Orders can have multiple items and sometimes multiple reviews. Directly joining orders, items and reviews multiplies rows. The implementation groups items to one row per order, separately summarises reviews per order, and then performs validated joins to retain exactly one fact record per order. All SLA percentages use this order-level fact table.

4.2 Seller attribution

When a delivered order contains products from more than one seller, delivery outcome cannot be fairly assigned to one seller. Seller-specific comparisons in the starter code therefore restrict to single-seller orders. Multi-seller records remain in the overall late-rate denominator.

4.3 Reporting layer

The Phase 1 reporting table is deliberately simple. In the later Power BI semantic model, it can be split into an order fact table and dimensions for purchase date, customer geography and unique seller; this follows Microsoft's published star-schema guidance.

5. KPI definitions and calculation rules

KPI	DEFINITION / FORMULA
Eligible delivered orders	Delivered status + present promised and actual timestamps + delivery not earlier than purchase
Late orders	Eligible orders where DATE(actual delivery) > DATE(estimated delivery)
Late delivery rate	Late orders / eligible delivered orders × 100%
On-time rate	Eligible, non-late orders / eligible delivered orders × 100%
Delay days	MAX(0, delivered calendar date − promised calendar date), for eligible records
Approval duration	Approval timestamp − purchase timestamp, if valid
Pre-handoff duration	Carrier-handoff timestamp − approval timestamp, if valid
In-transit duration	Customer delivery timestamp − carrier-handoff timestamp, if valid
Review score	Mean numeric score per order; reviewed-only comparisons explicitly labelled

Important clock-time edge case: when the promised timestamp is midnight, an order delivered at 20:30 on the promised calendar date is on time. Comparing full timestamps without normalising to dates would incorrectly classify such orders as late.

Denominator audit: canceled, undelivered or chronologically invalid records are excluded from SLA calculations and counted in a separate data-quality report, not silently discarded. Stage durations are null when timestamps are absent or reversed; they are never filled with zero.

6. Step-by-step analytical methodology

6.1 Data profiling and cleaning

1. Confirm file completeness and required column names; verify unique order/customer/seller keys and unique item-level composite keys.
2. Inspect status counts, missing timestamp patterns, invalid date parsing and chronological contradictions.
3. Aggregate review scores and item/seller associations to order grain; reject joins that change row count.
4. Build eligible_sla, is_late, late_days, stage durations and exclusion/quality flags.

6.2 Baseline and diagnostic SQL

1. Compute eligible order counts, late order counts, rates and monthly purchase-cohort trends.

2. Rank customer states by late rate using CASE, GROUP BY, HAVING, common table expressions and window functions, checking group sizes.
3. Compare single-seller order outcomes; always distinguish observed association from responsibility.
4. Reconcile SQL-derived totals against Python output and dashboard cards.

6.3 Stage and review investigation

For timestamps with valid ordering, compare median processing and transit durations between late and on-time groups. Present counts of missing or rejected stage records beside each result. Compare review means and low-score shares among reviewed orders; report how many eligible orders have no score.

6.4 Statistical robustness

Proposed extensions: Wilson intervals for rates; bootstrap uncertainty on review-score differences; stratification by purchase month and customer geography; alternative minimum-volume thresholds; sensitivity to excluding questionable timestamps. P-values alone do not establish practical significance or causation.

6.5 Recommendations process

Translate verified patterns into proposed tests. Each recommendation must name a target segment, observed supporting result, plausible competing explanations, measurable next action, monitoring KPI and owner. If evidence is weak, recommend improved instrumentation rather than a confident operational change.

7. Power BI dashboard specification

PAGE	COMPONENTS	MANAGEMENT QUESTION
Executive overview	KPI cards, late-rate trend, delivery volumes, segment filter	Are we meeting delivery promises?
Fulfilment stages	Duration distributions, stage quality notes, subgroup matrix	At which recorded stage is time accumulating?
Customer experience	Reviews vs lateness, score distributions, review coverage	How is lateness associated with satisfaction?
Action plan	Prioritised investigation, caveats, follow-up KPIs	Where should we investigate or trial changes first?

Power BI measures must use the same eligible denominator as the Python and SQL layers. Rate measures should use DIVIDE and be formatted as percentages. Avoid confusing purchase month with delivery month. Label the report “historical 2016–2018 Olist data,” and do not imply live performance.

8. Learning roadmap (14-day sprint)

This two-week sprint requires approximately 3–5 focused hours per day, and prioritises a credible investigation over extra technology. Deliver the validated SQL/Python analysis, a 2–3-page Power BI report, and a management memo. Avoid building predictive ML, live apps or cloud scheduling before the core project is finished.

DAY	DELIVERABLE	WHAT YOU LEARN
1	Environment, tests, demo and official data	VS Code, virtualenv, source provenance
2	Data model and quality profile	pandas, keys, one-to-many joins, missingness

DAY	DELIVERABLE	WHAT YOU LEARN
3	Audited real order-level fact table	Cleaning, timestamp rules, grain validation
4	Verified late-delivery baseline	SQL CTEs, CASE, denominator reconciliation
5	Monthly trend and delay distributions	Date grouping, outliers, descriptive analysis
6	Fulfilment stage comparisons	Durations, medians, null/invalid sequences
7	Interim findings log and research focus	Prioritisation, communication, evidence gaps
8	State and single-seller investigation	Segmentation, sample-size safeguards
9	Reviews vs delivery timeliness	Associations, coverage, selection bias
10	Two robustness/sensitivity checks	Statistics, confounding, claim validation
11	Power BI model + reconciled measures	Power Query, DAX, filters and relationships
12	2–3-page management dashboard	Visualization, interactive reporting
13	Findings and action recommendations	Business reasoning, evidence-to-action writing
14	QA, GitHub and interview walkthrough	Reproducibility, final checks, presentation

8.1 Sprint review gates

- Day 2: all official input files acquired; source schema and row-grain clearly described.
- Day 4: eligibility and late-delivery baseline independently agree in Python and SQL.
- Day 8: stage and segment results include sample sizes and exclusions, not just rankings.
- Day 12: Power BI results reconcile with the trusted tables and report clearly states historical data coverage.
- Day 14: every conclusion links to a validated query/figure, with an alternative explanation and a sensible business next step.

9. Testing, quality assurance and acceptance criteria

TEST	PASS CRITERION
Row grain and key uniqueness	Fact row count equals orders row count; no duplicate order_id
Date boundary test	Delivery on promised date counts as on time even if after midnight
Ineligible status test	Canceled, missing or impossible delivery chronology excluded
Many-to-many prevention	Multi-review and multi-item orders still count once
Multi-seller safeguard	Multi-seller order never assigned to single_seller_id
Report reconciliation	Python, SQL and Power BI metrics match same filters
Audit transparency	Excluded rows, invalid stages and missing reviews reported
Communication integrity	Historical data and causal limits prominently displayed

The included starter repository implements four automated pytest checks using a tiny synthetic fixture, and those tests passed in the creation environment. Passing synthetic tests does not replace auditing the original Olist files.

10. Risks, constraints and mitigations

RISK	MITIGATION
Historical data mistaken for current conditions	Visible period/source label on report and CV
Raw multi-table joins inflate counts	One-order fact + validated merge keys
Incomplete or reversed timestamps	Null invalid stages; independent quality audit
Small groups produce unstable rankings	Volume filters and uncertainty intervals
Unobserved carrier/weather/geography effects	Non-causal language, competing hypotheses
Review selection bias	Compare coverage and reviewed-only denominators
Overclaiming “money saved”	No financial savings unless costs and assumptions are documented
Dataset rights and privacy	Cite source; inspect license; avoid publishing raw private data

11. Deliverables and completion rubric

ASSET	CURRENT STATUS	COMPLETION REQUIREMENT
Project charter / roadmap	Included in this report	Revise with actual decisions
Python analytics pipeline	Implemented starter	Run full Olist, profile exceptions
Data quality and KPI exports	Implemented starter	Record actual metrics in report
SQL investigations	Included	Reconcile to Python
Power BI specification	Included	Create and validate .pbix
Business findings report	Design complete; findings pending	Insert real evidence and limitations
GitHub portfolio case study	Starter README included	Screenshots, lessons, reproducible setup

11.1 Final evidence template

Finding [F1]: [Observed data result with population and period]. Evidence: [SQL query / chart / output file]. Interpretation: [Why the association matters]. Alternative explanation: [Potential confound]. Proposed action: [Targeted investigation / controlled trial]. Monitoring KPI: [Measure and review window]. Confidence: [Data and assumptions].

Do not fill this with invented percentages; calculate, check and then write the evidence. State whenever a recommendation is an untested hypothesis.

12. Immediate next actions

1. Unzip Delivery_Investigation_14_Day_Starter_Kit.zip and open the folder in VS Code.
2. Create a virtual environment, install the package and run pytest. Confirm four tests pass.
3. Execute the synthetic demonstration. Treat its metrics as test-only, not business evidence.
4. Download official Olist data from Kaggle; place the five named CSV files in data/raw/.
5. Run the real pipeline and inspect data_quality.json before interpreting kpis.json.

6. Explain why the late-rate denominator excludes canceled and chronologically impossible records.
7. Reproduce the baseline using sql/analysis_queries.sql in SQLite; compare counts and percentages.
8. Record your first genuine observation and a question it raises. Then move to geographic and stage analysis.

Appendix A. File structure

README.md — setup and evidence rules; docs/learning_roadmap.md — 14-day sprint plan; docs/days_01_02_workshop.md — setup workshop; docs/data_dictionary.md — fields and metrics; docs/power_bi_blueprint.md — planned visuals and DAX; src/delivery_investigation/ — Python transformation and analysis code; sql/analysis_queries.sql — auditable SQL; tests/ — edge-case checks; data/sample/ — synthetic demo; data/raw/ — local official Kaggle CSVs; outputs/ — generated results (ignored by Git).

Appendix B. Reporting metrics checklist

- Baseline: eligible delivered orders, late count, on-time count, late rate and delay days.
- Segmentation: purchase month, customer state and single-seller group; always show denominators.
- Process: approval, pre-handoff and transit durations with valid-observation counts.
- Experience: review coverage, mean score and share rated 1–2 in both delivery groups.
- QA: source row count, exclusion reasons, invalid timestamp sequences, missing geographic/seller matches.

Appendix C. Portfolio interview questions

- Why did you measure calendar-date lateness rather than compare raw timestamps?
- How did you prevent item and review joins from inflating the number of orders?
- What evidence is needed before claiming sellers or couriers caused delivery delays?
- How might review-response bias affect the satisfaction result?
- How would you improve this investigation if the business gave you tracking events and carrier IDs?

References and further reading

1. Olist and collaborators. Brazilian E-Commerce Public Dataset (2016–2018). Kaggle. https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

2. Microsoft Learn. Understand star schema and the importance for Power BI. https://learn.microsoft.com/en-us/power-bi/guidance/star-schema

3. pandas documentation. pandas.to_datetime. https://pandas.pydata.org/docs/reference/api/pandas.to_datetime.html

4. Olist project repository README and code (starter kit packaged alongside this report).